

NETWORKS AND COMPLEXITY

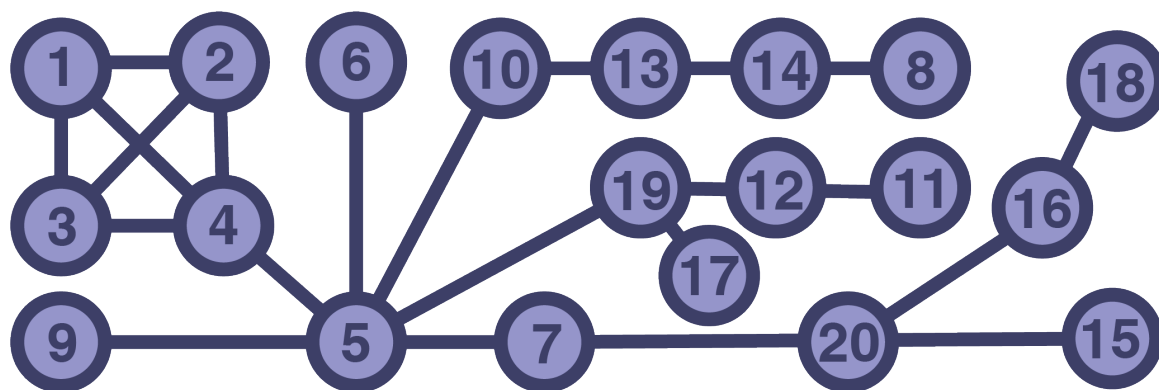
Solution 20-12

*This is an example solution from the forthcoming book *Networks and Complexity*.*

Find more exercises at <https://github.com/NC-Book/NCB>

Ex 20.12: A large eigenvalue puzzle [Lvl. 5]

Find as many eigenvalues as possible of the dragon network:



Solution

This is an unsolved puzzle in the sense that people may still come up with clever ways to spot eigenvalues that I wasn't able to find. However, there is an eigenvalue of 0 and two of -1 that are in plain sight. Another zero is quite hidden, and it wouldn't be fun if there wasn't a golden ratio to find as well. Maybe you can even find some more.

One of the benefits of doing network math is that you don't need to do Sudokus anymore. You can just draw a network like this and have hours of fun finding the eigenvalues.